ITA0605 - Machine Learning
List of Lab Exercises

18. To classify Iris flowers using the Perceptron algorithm.

Aim:

To classify Iris flowers into different species using the Perceptron algorithm.

Algorithm:

1. Import the required Python libraries and Iris dataset.
2. Separate the input features and target classes.
3. Split the dataset into training and testing sets.
4. Create and train the Perceptron model using the training data.
5. Predict the classes of the test data.
6. Calculate the accuracy of the Perceptron model.
7. Display the classification results and accuracy.

Dataset 1:

Program Code:

```
import pandas as pd
from sklearn.linear_model import Perceptron
from sklearn.preprocessing import StandardScaler
data = {
    'Weight': [150, 160, 145, 120, 125, 130, 155, 118, 148, 122],
    'Diameter': [7.5, 7.8, 7.3, 5.5, 5.8, 6.0, 7.6, 5.4, 7.4, 5.7],
    'Sweetness': [90, 88, 91, 70, 72, 74, 89, 69, 92, 71],
    'Fruit': ['Apple', 'Apple', 'Apple', 'Orange', 'Orange',
              'Orange', 'Apple', 'Orange', 'Apple', 'Orange']
}

df = pd.DataFrame(data)

X = df[['Weight', 'Diameter', 'Sweetness']]
```

y = df['Fruit']

scaler = StandardScaler()

X_scaled = scaler.fit_transform(X)

model = Perceptron(random_state=42)

model.fit(X_scaled, y)

new_sample = [[152, 7.5, 90]]

new_sample_scaled = scaler.transform(new_sample)

prediction = model.predict(new_sample_scaled)

print("Predicted Fruit:", prediction[0])

Output:

```
# Train Perceptron
model = Perceptron(random_state=42)
model.fit(X_scaled, y)

# New sample
new_sample = [[152, 7.5, 90]]
new_sample_scaled = scaler.transform(new_sample)

# Prediction
prediction = model.predict(new_sample_scaled)

print("Predicted Fruit:", prediction[0])
```

```
... Predicted Fruit: Apple
```

Dataset 2:

Program Code:

import pandas as pd

from sklearn.linear_model import Perceptron

from sklearn.preprocessing import StandardScaler

data = {

'CGPA': [9.3, 8.9, 8.6, 7.3, 6.9, 9.1, 7.4, 8.7, 6.6, 8.8],

'Aptitude': [92, 87, 84, 74, 69, 91, 75, 85, 64, 88],

'Communication': [90, 85, 82, 72, 67, 89, 73, 83, 60, 86],

'Placement': ['Placed', 'Placed', 'Placed', 'Not Placed', 'Not Placed',

```
        'Placed', 'Not Placed', 'Placed', 'Not Placed', 'Placed']
}

df = pd.DataFrame(data)

X = df[['CGPA', 'Aptitude', 'Communication']]
y = df['Placement']
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
model = Perceptron(random_state=42)
model.fit(X_scaled, y)
new_sample = [[8.8, 86, 84]]
new_sample_scaled = scaler.transform(new_sample)
prediction = model.predict(new_sample_scaled)
print("Predicted Placement:", prediction[0])
```

Output:

```
# Train Perceptron
model = Perceptron(random_state=42)
model.fit(X_scaled, y)

# New sample
new_sample = [[8.8, 86, 84]]
new_sample_scaled = scaler.transform(new_sample)

# Prediction
prediction = model.predict(new_sample_scaled)

print("Predicted Placement:", prediction[0])
```

```
... Predicted Placement: Placed
```

Dataset 3:

Program Code:

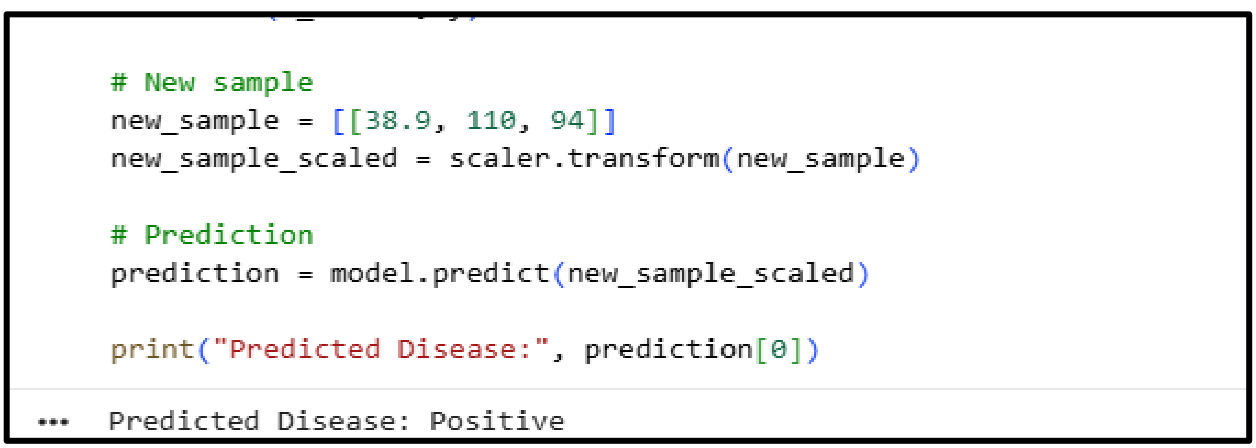
```
import pandas as pd
from sklearn.linear_model import Perceptron
from sklearn.preprocessing import StandardScaler
```

```
data = {
    'Temperature': [39.2, 38.8, 37.0, 39.1, 36.8, 38.9, 37.2, 39.3, 36.7, 38.7],
    'HeartRate': [112, 110, 78, 114, 76, 111, 80, 115, 74, 109],
    'OxygenLevel': [93, 94, 99, 92, 98, 94, 99, 92, 98, 95],
    'Disease': ['Positive', 'Positive', 'Negative', 'Positive', 'Negative',
                'Positive', 'Negative', 'Positive', 'Negative', 'Positive']
}

df = pd.DataFrame(data)

X = df[['Temperature', 'HeartRate', 'OxygenLevel']]
y = df['Disease']
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
model = Perceptron(random_state=42)
model.fit(X_scaled, y)
new_sample = [[38.9, 110, 94]]
new_sample_scaled = scaler.transform(new_sample)
prediction = model.predict(new_sample_scaled)
print("Predicted Disease:", prediction[0])
```

Output:



Dataset 4:

Program Code:

```
import pandas as pd
from sklearn.linear_model import Perceptron
from sklearn.preprocessing import StandardScaler
data = {
    'Experience': [10, 9, 8, 4, 3, 11, 5, 8, 2, 7],
    'Performance': [95, 92, 90, 72, 68, 96, 75, 89, 60, 87],
    'TrainingHours': [50, 48, 45, 28, 24, 52, 30, 44, 18, 40],
    'Promotion': ['Promoted', 'Promoted', 'Promoted', 'Not Promoted',
                  'Not Promoted', 'Promoted', 'Not Promoted', 'Promoted',
                  'Not Promoted', 'Promoted']
}

df = pd.DataFrame(data)

X = df[['Experience', 'Performance', 'TrainingHours']]
y = df['Promotion']
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
model = Perceptron(random_state=42)
model.fit(X_scaled, y)
new_sample = [[8, 90, 46]]
new_sample_scaled = scaler.transform(new_sample)
prediction = model.predict(new_sample_scaled)
print("Predicted Promotion:", prediction[0])
```

Output:

```
# New sample
new_sample = [[8, 90, 46]]
new_sample_scaled = scaler.transform(new_sample)

# Prediction
prediction = model.predict(new_sample_scaled)

print("Predicted Promotion:", prediction[0])
```

```
••• Predicted Promotion: Promoted
```

Dataset 5:

Program Code:

```
import pandas as pd
from sklearn.linear_model import Perceptron
from sklearn.preprocessing import StandardScaler
data = {
    'EngineCapacity': [100, 125, 150, 800, 1000, 1200, 110, 900, 140, 1300],
    'Weight': [95, 110, 125, 850, 950, 1050, 100, 900, 120, 1100],
    'Mileage': [70, 65, 55, 22, 20, 18, 68, 21, 58, 17],
    'VehicleType': ['Bike', 'Bike', 'Bike', 'Car', 'Car', 'Car',
                    'Bike', 'Car', 'Bike', 'Car']
}

df = pd.DataFrame(data)

X = df[['EngineCapacity', 'Weight', 'Mileage']]
y = df['VehicleType']
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
model = Perceptron(random_state=42)
model.fit(X_scaled, y)
new_sample = [[135, 118, 60]]
new_sample_scaled = scaler.transform(new_sample)
```

```
prediction = model.predict(new_sample_scaled)
print("Predicted Vehicle Type:", prediction[0])
```

Output:

```
# New sample
new_sample = [[135, 118, 60]]
new_sample_scaled = scaler.transform(new_sample)

# Prediction
prediction = model.predict(new_sample_scaled)

print("Predicted Vehicle Type:", prediction[0])
```

```
••• Predicted Vehicle Type: Bike
```

Result:

The Perceptron algorithm was successfully implemented on all five classification datasets. The model successfully predicted the class of each given new sample. Thus, classification was successfully performed using the Perceptron algorithm.